THE RESTRICTION OF THE RELATIVE DUALIZING COMPLEX TO A FIBRE IS A DUALIZING COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0BZZ (sources/stacks-duality/pinned-original-source.tex line 5130).

Allowed standard prerequisites (editorial).

- Cohomological boundedness of the relative shriek for perfect morphisms. Same chapter, Lemma labelled lemma-perfect-comparison-shriek, cited at proof lines 5139-5141. Places omega^bullet_{X/Y} in D^b_Coh.
- Local base change for the shriek. Same chapter, Lemma labelled lemma-base-change-locally, cited at proof lines 5141-5144. Reduces to the case of the zero dimensional base.
- The shriek of a dualizing complex is dualizing. Same chapter, Lemma labelled lemma-shriek-dualizing, cited at proof lines 5144-5147. Gives the conclusion for the fibre.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6U: sources/stacks-duality/pinned-original-source.tex, source lines 4798–4798. Same-chapter pointer into the author environment lemma at source line 4798; invoked at proof line 5141 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BZY: sources/stacks-duality/pinned-original-source.tex, source lines 5094–5094. Same-chapter pointer into the author environment lemma at source line 5094; invoked at proof line 5143 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA3: sources/stacks-duality/pinned-original-source.tex, source lines 4708–4708. Same-chapter pointer into the author environment lemma at source line 4708; invoked at proof line 5146 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. Assume f is flat. Set $\omega^\bullet_{X/Y} = f^!\mathcal{O}_Y$ in $D^b_{Coh}(X)$. Let $y \in Y$ and $h : X_y \to X$ the projection. Then $Lh^*\omega^\bullet_{X/Y}$ is a dualizing complex on X_y.*

Proof. The complex $\omega^\bullet_{X/Y}$ is in D^b_{Coh} by Lemma 0B6U. Being a dualizing complex is a local property. Hence by Lemma 0BZY it suffices to show that $(X_y \to y)^!\mathcal{O}_y$ is a dualizing complex on X_y. This follows from Lemma 0AA3. □